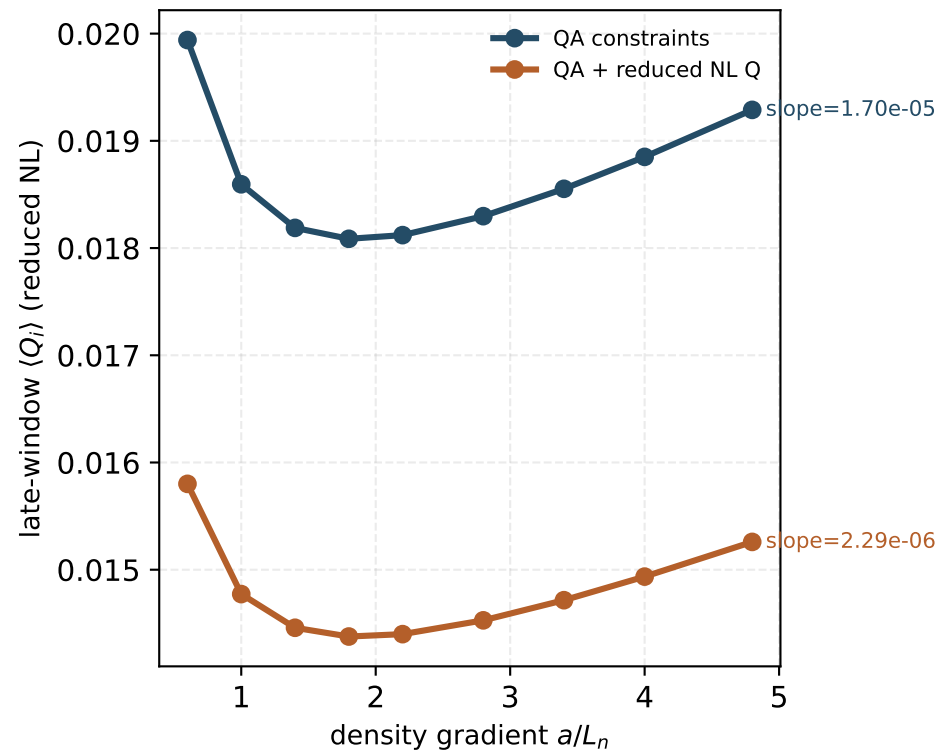
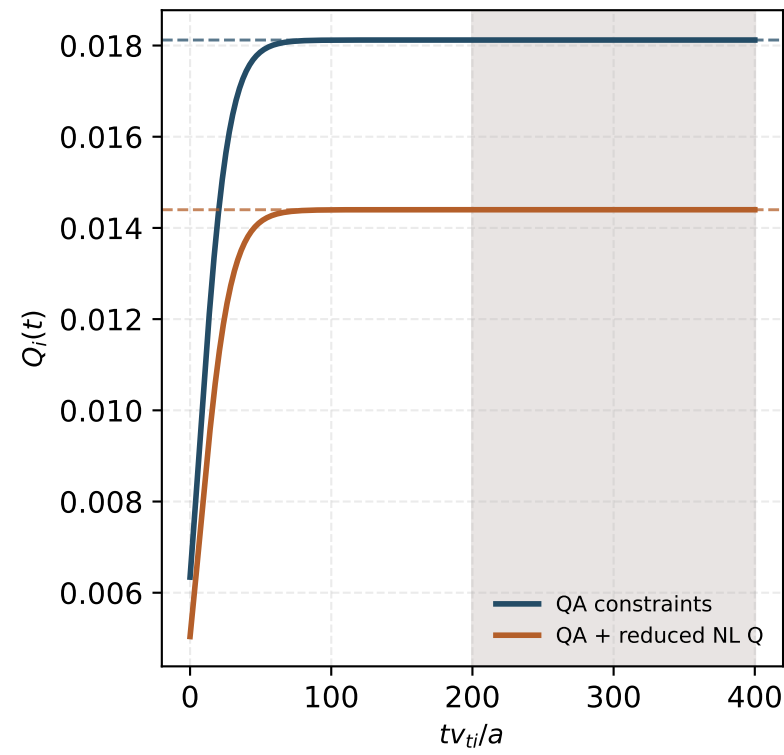


Aspect-6 QA low-turbulence optimization: constraints-only vs transport-aware design

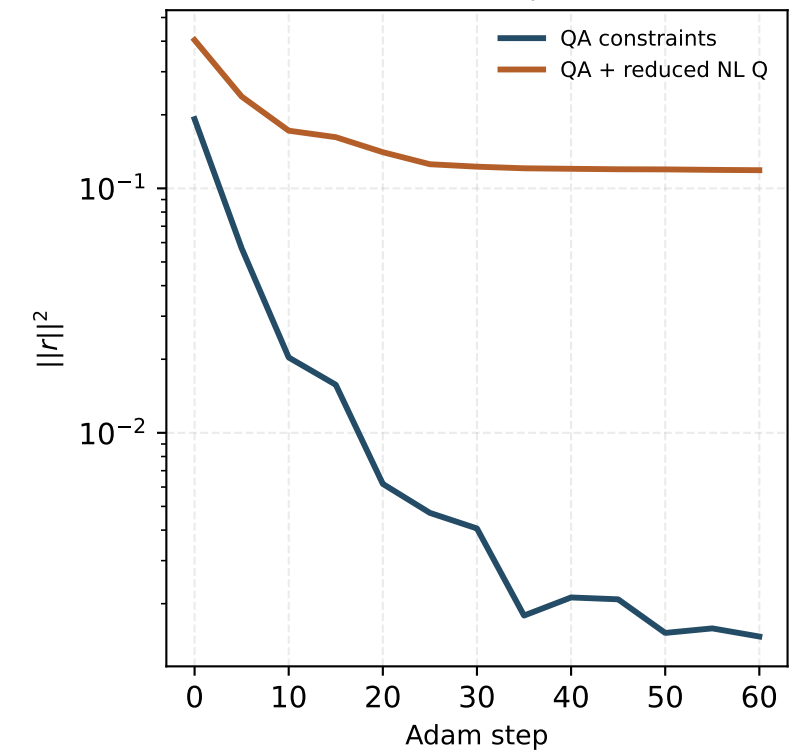
Gradient scan at fixed $a/L_{Ti} = 6$



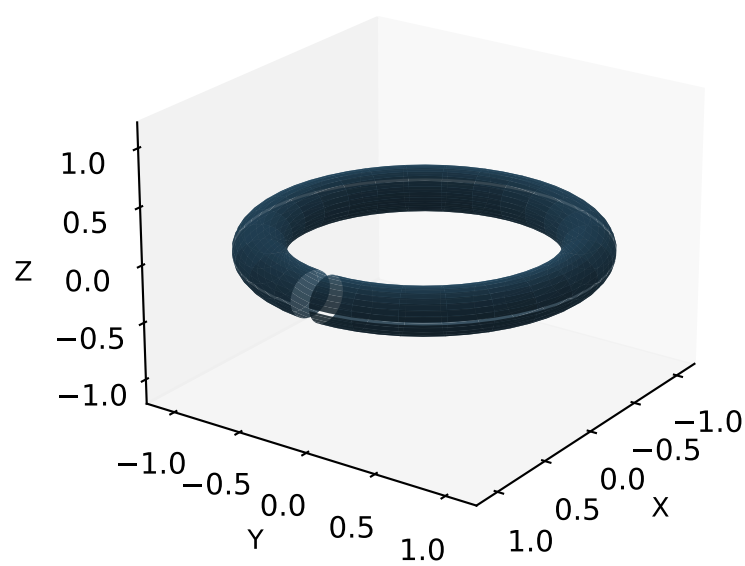
Fixed-gradient trace: $a/L_n = 2.2$, $a/L_{Ti} = 6.0$



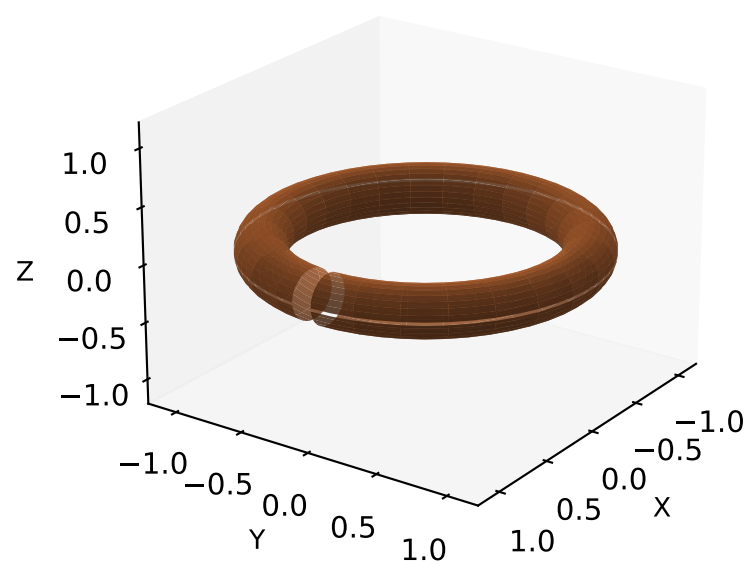
Constrained objectives



Reduced LCFS: QA constraints



Reduced LCFS: QA + reduced NL Q



Gate summary

- Aspect target: 6.0
- Minimum iota: 0.41
- Operating iota floor: 0.70
- AD/FD gates passed: True
- Constraint gates passed: True
- Long-window gates passed: True
- Fixed-gradient Q reduction: 20.5%
- Trace tmax: 400

Final values

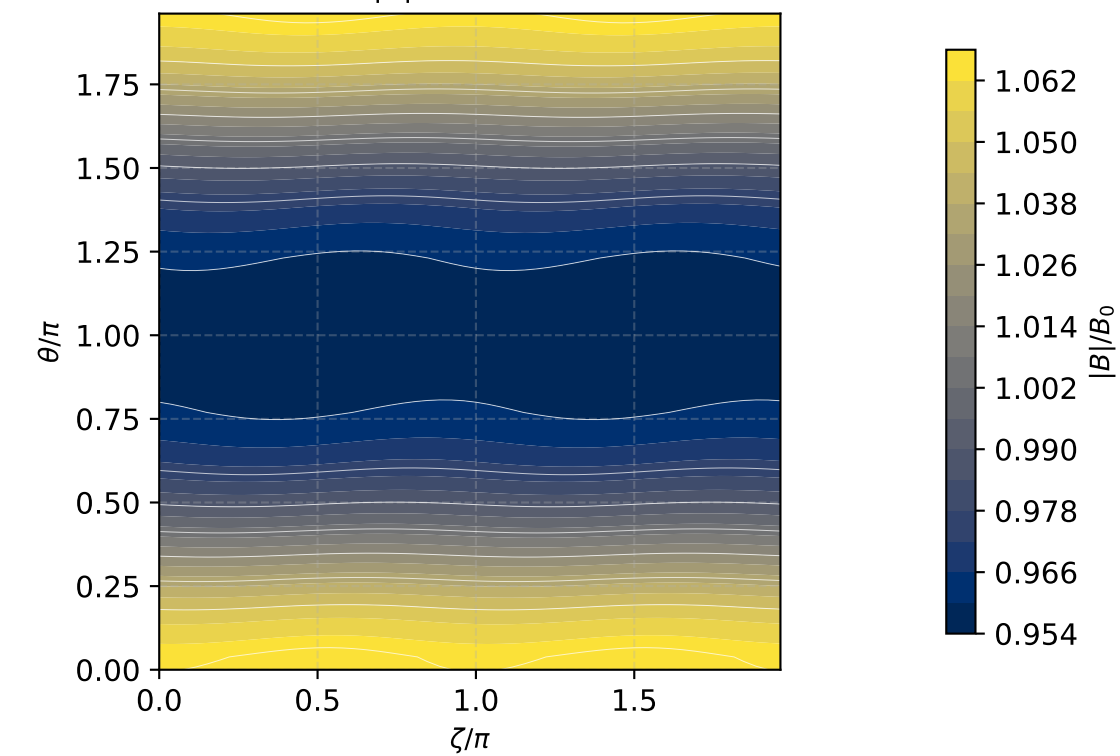
QA constraints

- A=5.986, iota=0.739
- QA residual=1.99e-03
- $\langle Q_i \rangle = 1.812e-02$

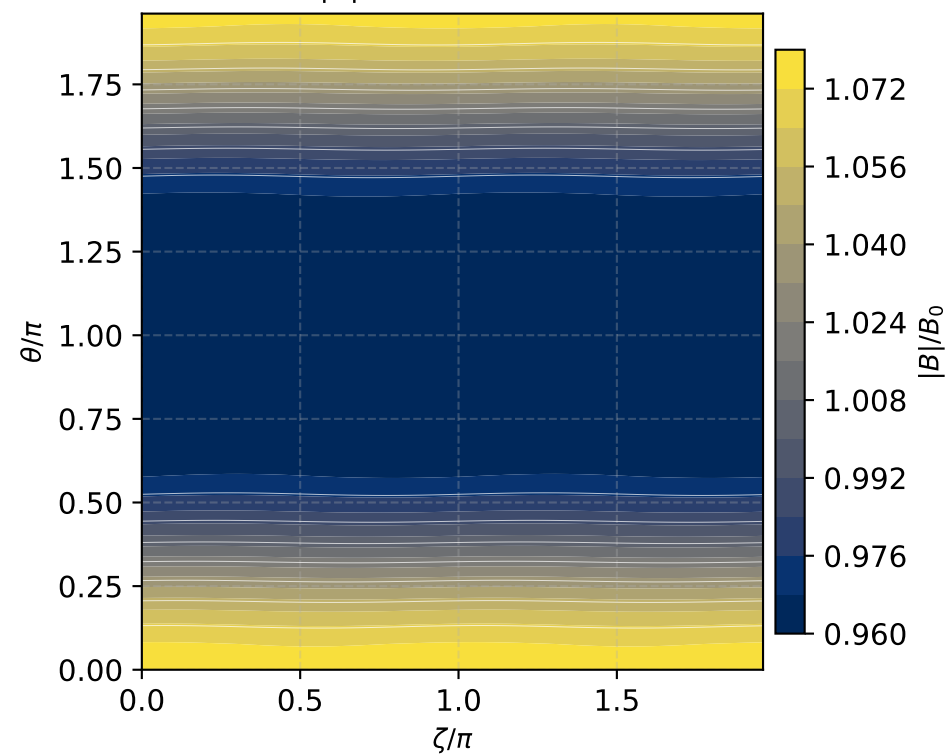
QA + reduced NL Q

- A=5.976, iota=0.756
- QA residual=1.01e-03
- $\langle Q_i \rangle = 1.440e-02$

LCFS $|B|$: QA constraints



LCFS $|B|$: QA + reduced NL Q



Scope and method

- Reduced max-mode-1 QA controls
- Aspect-ratio and two iota-floor constraints
- Long differentiable RK2 nonlinear ITG envelope
- Scalar/residual/observable AD-FD gates
- Production nonlinear claims still need long-window GK audits

Equations used in this panel

$J = \|r_A, r_i, r_{QA}, r_{reg}, r_Q\|^2$

$dE/dt = 2\gamma E - \alpha E^2$

$Q_i = W_i E$